

their expertise, whereas the one that defeated it had content that could be edited by anyone with internet connectivity.

What is the name of the winner? Of course, it is Wikipedia. But how did this happen? How did Wikipedia become the ultimate source of knowledge in the world? It was the collective power of selfless human effort. Indeed, on many occasions, I have come across Wikipedia articles edited by Nobel laureates and Fields Medalists.

Another remarkable area where people collaborate selflessly is software development. Consider all the free and open source software available to individuals across various fields and aspects of life. Two things constantly amaze me about free and open source software. First, the exceptional dedication of numerous talented individuals who spend their leisure time on complex programming so that all of humanity can benefit. Second, the incredible range of free and open source software, from simple text editors like Nano and Atom to sophisticated operating system distributions like Ubuntu, Fedora, and Debian. This broad spectrum of software has also led to the creation of tools that are useful to even very small, specific groups of people. Moreover, software has been developed for communities that might not otherwise afford the cost of software development. While there are countless free and open source programs, our focus in this article is on software designed for special education, aimed at teaching and training individuals with neurodevelopmental disabilities such as Down syndrome, autism, and cerebral palsy.

Given the relatively low general awareness regarding these distinct conditions, all of which result in different challenges and require unique training and treatment,

let us discuss them briefly before we proceed any further. Down syndrome is a genetic disorder caused by the presence of an extra copy of chromosome 21, leading to developmental and intellectual delays as well as characteristic physical features. Autism, also known as autism spectrum disorder, is a neurodevelopmental disorder characterised by difficulty in social interaction and repetitive behavioural patterns. Cerebral palsy is another distinct condition, comprising a group of movement disorders that appear in early childhood, caused by abnormal development or damage to the parts of the brain that control movement, balance, and posture. Symptoms include poor coordination, weak muscles, and tremors. I urge all readers to consult Wikipedia articles on these three conditions to learn and spread awareness among others. This would be a great service to humanity, as the training and rehabilitation necessary for these individuals are not yet adequate in India, although standards are improving.

Many individuals with Down syndrome, autism, and cerebral palsy face challenges in language formation, experiencing difficulties ranging from speaking to understanding language. These challenges significantly impact communication abilities, making the professional services of a speech-language pathologist (SLP) essential for assessment, diagnosis, and training. However, many special schools for children with Down syndrome, autism, and cerebral palsy lack the services of a full-time professional SLP due to financial constraints and a shortage of qualified professionals. This situation necessitates the use of specialised software support. In this article, we will discuss software that can help affected individuals to some extent. While the services of an SLP are ideal, the software we discuss here serves as supplementary support.

Augmentative and alternative communication (AAC) devices

A number of devices called augmentative and alternative communication (AAC) devices are used to provide communication methods that supplement or replace speech or writing for people having difficulties with spoken or written language. AAC devices can help individuals with birth conditions like Down syndrome, autism, and cerebral palsy to communicate effectively. They are also helpful to people suffering from diseases such as Parkinson's, which can affect individuals later in life. There are two types of AAC devices: low-tech and high-tech. Low-tech AAC devices include communication boards, picture exchange communication systems, eye gaze boards, and so on, which are priced under a few thousand rupees. High-tech AAC devices include speech-generating devices, tablets with AAC apps, and dedicated communication devices, with prices ranging from a few thousand rupees to millions of rupees.

When I first learned about AAC devices used for training in many developed countries, I was saddened to recall that none of the ten or so special schools I visited in the past two decades had anything close to these devices, despite the availability of inexpensive low-tech AAC devices in the market. All they had were charts with pictures of fruits, animals, etc, hanging in the classrooms. Furthermore, the computers (at most three, often slow desktops) they had were used mostly for photo ops. I saw photos of students sitting in front of computers, looking confused at their names displayed in large font in Microsoft Word, typed by someone else. However, this lack of technology in special education could be easily overcome with free and open source software, as many such programs can emulate the specialised hardware of an AAC device on even low-end laptops. The adoption of such software would transform special education in India.